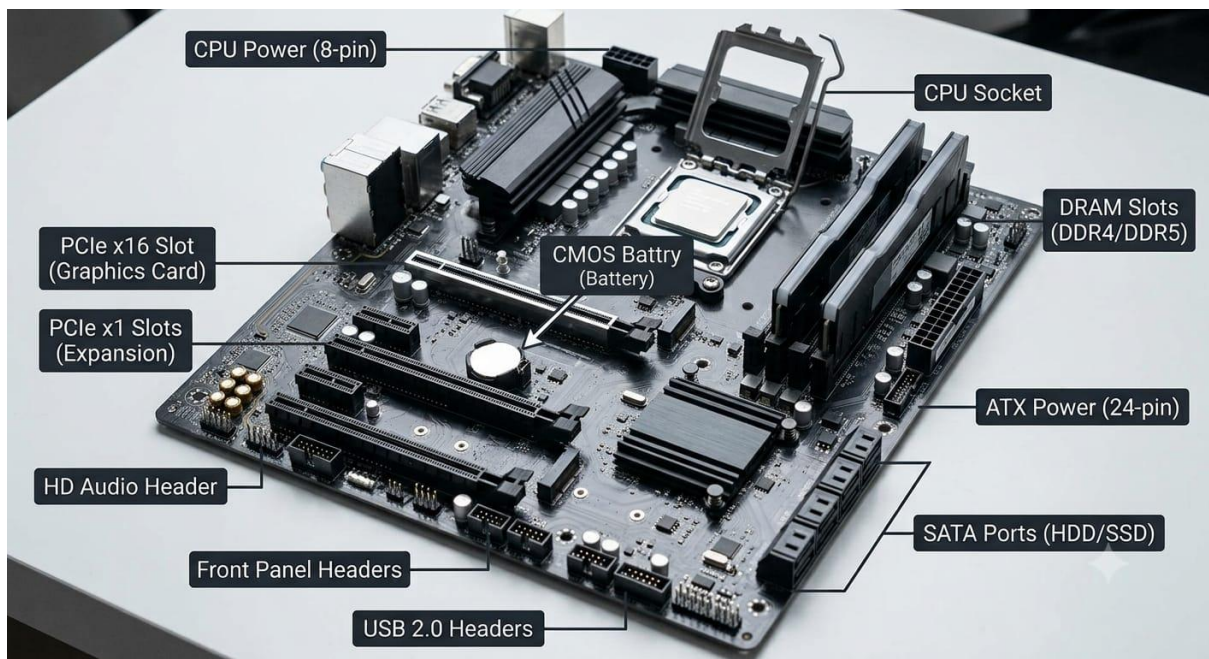


ASSIGNMENT 1.2

1. Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



2. Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

(1) LGA 1151

- **Compatible CPU:** Intel core i7-6700K or i9-9900K
- **Supported Generation:** 6th Gen, 7th Gen or 8th Gen, 9th Gen

(2) AM4

- **Compatible CPU:** AMD Ryzen 5 5600X
- **Overview:** AMD supporting Ryzen 1000 through 5000 series processors

(3) LGA 1200

- **Compatible CPU:** Intel Core i9-10900K
- **Supported Generation:** 10th & 11th Gen

3.Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

1) Primary Functions of the CPU and Chipset:

CPU (Central Processing Unit)

The CPU is the “brain” of the computer.

- **Processing and calculations:** it performs all mathematics and logical tasks of the system.
- **Execution of instructions:** the CPU’s job is to understand and execute all directions/instructions received from software or applications.

Chipset

The chipset is the “communication manager” on the motherboard.

- **Data flow control:** It manages the exchange of data between the CPU, RAM, storage (SSD/HDD), graphics card, and input/output devices like USBs.
- **Connectivity:** The chipset connects the CPU with other computer components and securely delivers information between them.

2) How do they work together?

When you open Spotify on a computer, both these components work step-by-step:

1. Accepting User Input:

When you click on Spotify using the mouse, the chipset receives the click signal from the mouse and sends it to the CPU.

2. Loading data from storage:

The CPU realizes that Spotify needs to be launched, so it instructs the chipset to transfer Spotify’s data from the storage drive into RAM.

The chipset provides the pathway for data transfer between the SSD and RAM.

3. Processing the App:

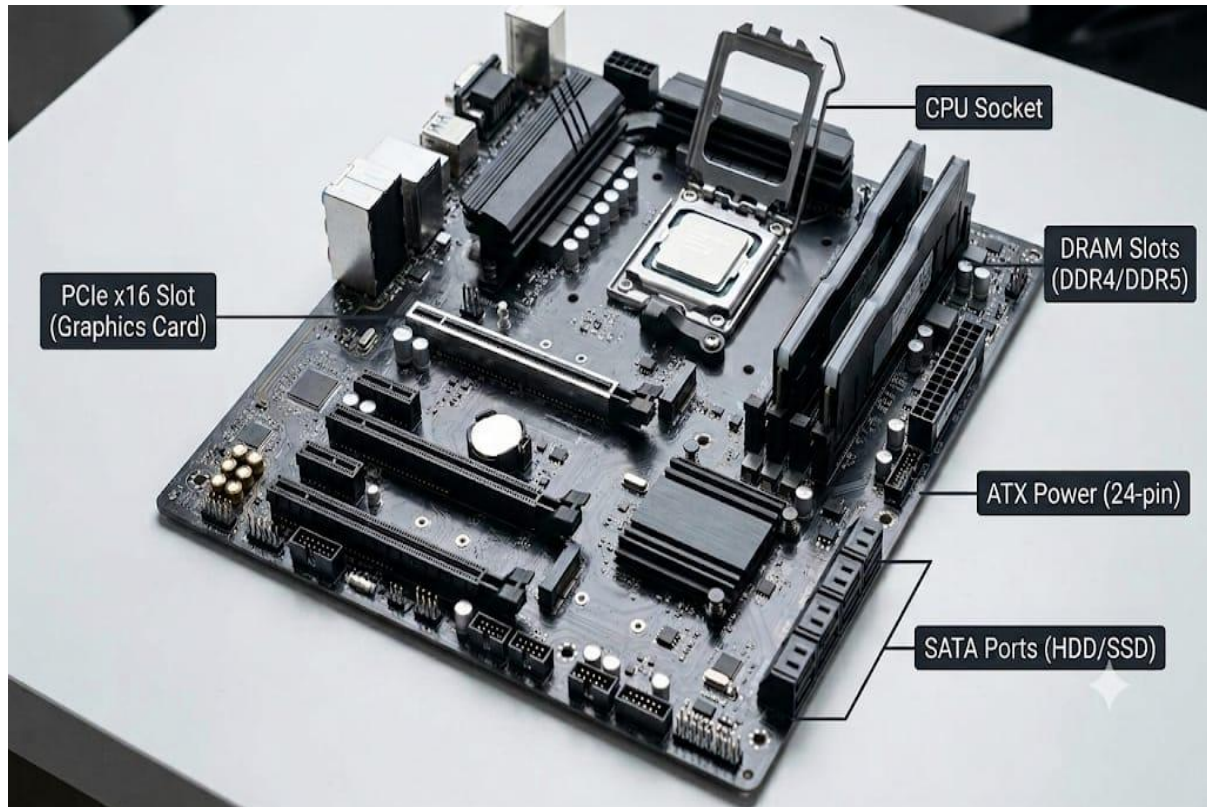
Once the data arrives in RAM, the CPU processes Spotify’s code and opens the app’s user interface (UI) on the screen.

4. Audio and Output Management:

When you play a song, the CPU decodes the music file.

The CPU then sends those audio signals through the chipset to the sound card/speakers and the visual signals to the monitor.

4. Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labelling each one.



1. CPU socket

- **Location:** Top-center area of the motherboard.
- **Visual Description:** A large, square socket with a small metal latch arm next to it.
- **Function:** Holds the CPU.

2. DRAM slots:

- **Location:** to the right of the CPU socket.
- **Visual description:** long vertical slots with plastic retention clips at the ends.
- **Function:** holds the computer's RAM sticks.

3. PCIe x16 Expansion slot

- **Location:** Below the CPU socket.
- **Visual Description:** The longest horizontal slot, often featuring a small plastic latch at the right end.
- **Function:** Holds dedicated graphics cards or expansion cards.

4. 24-pin ATX main power connector

- **Location:** Along the far right edge of the motherboard.
- **Visual Description:** A long rectangular plastic block containing 24 pin holes.
- **Function:** Receives main power from the power supply unit (PSU).

5. SATA ports

- **Location:** located at the bottom-right corner.
- **Visual Description:** small rectangular L-shaped ports arranged in a grid.
- **Function:** connects storage devices like SSDs and hard Drives.